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Data Modelling and Normal Forms

RDB | MTD.ba VZ SS26

Hagenberg 25.03.2026

HAGENBERG | LINZ | STEYR | WELS

ANSI-SPARC Architecture

External Level

- Represents the *view of the user*
- Only information relevant to specific user
- Excludes irrelevant and unauthorized data

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- Independent of hardware and software

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- *Data model* for entire data
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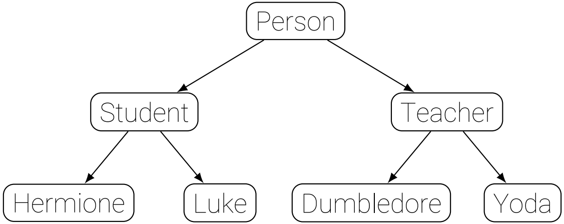
Internal Level

- *Physical representation* of database on computer system.
- How the data is stored

Types of data models

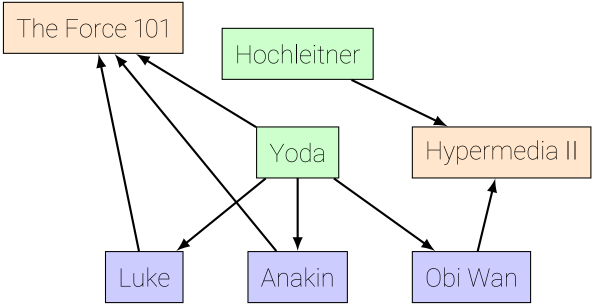
- Hierarchical model
- Network model
- relational model
- Object-oriented model
- Object-relational model
- Entity-Relationship (ER) model

Hierarchical model



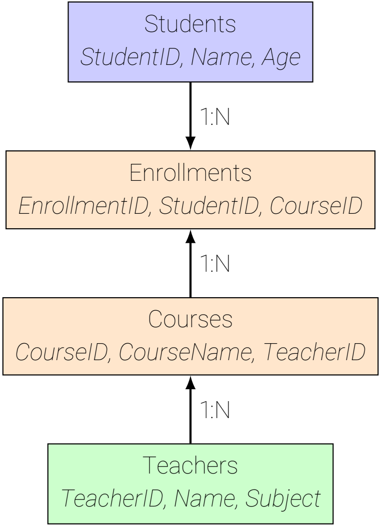
- Tree with a root node
- Every Branch has one parent

Network model



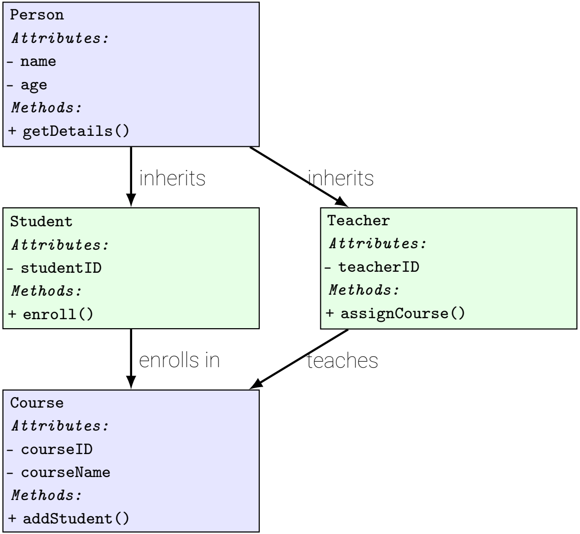
- Relationships between nodes
- Data represented as graph
- Relationships between multiple nodes possible

Relational model



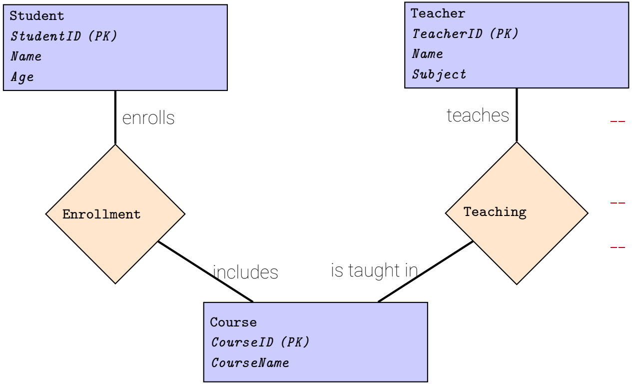
- Arranges data into tables
- Tables consist of columns and rows
- Relationships between tables

Object-relational model



- Object defines attributes and methods
- Objects can inherit from each other
- Relationships to other objects can be defined

Entity-Relationship (ER) model



- Defines real-world entities and relationships between them
- Relationships modeled separately
- Multiple entities can connect via one relationship

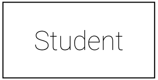
ER Models

Why ER Modelling?

- Define **entities** (e.g. tables)
- Define **attributes** (e.g. columns)
- Identify **relationships** between entities (foreign keys)
- Identify **cardinalities**

ER Model definitions

Entity



Is a real world object represented in the data model(e.g. Class, Table, ...).
Examples: Student, Teacher, Course, . . .

ER Model definitions

Entity

Student

Is a real world object represented in the data model(e.g. Class, Table, ...). Examples: Student, Teacher, Course, . . .

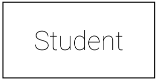
Attribute

Name

Is an attribute (column, field, ...) of an entity holding information Examples: Name, Age, Gender, Grade, . . .

ER Model definitions

Entity



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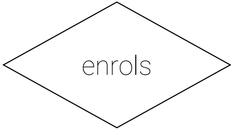
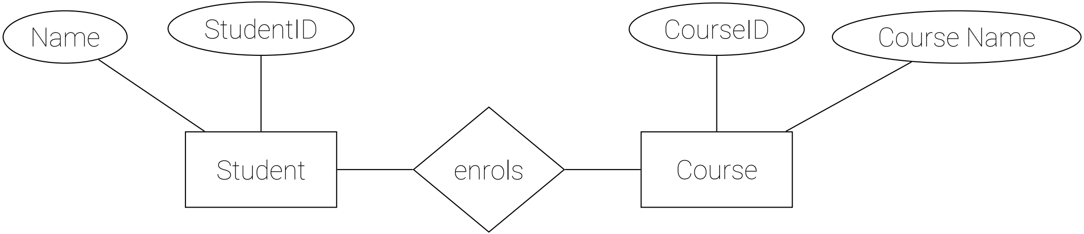
Relationship



Connects one or more entities, and defines how they are related with each other. Examples: enrols (Student to Course), teaches (Teacher to Course) . . .

Example


Name
StudentID
CourseID
Course Name
Student
enrols
Course

Attribute Types

Attribute

Name

Is an attribute (column, field, ...) of an entity holding information Examples: Name, Age, Gender, Grade

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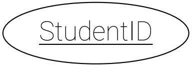
Key attribute



Uniquely identifies a specific entity. Examples: ID, Social Security Number, URL

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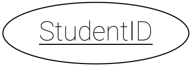

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Multivalued attribute

Attribute consisting of multiple values (for Database needs to be split into columns or own table) Examples: Address, Name (First, Last, Title, . . .),

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Attribute

Is an attribute (column, field, ...) of an entity holding information Examples: Name, Age, Gender, Grade

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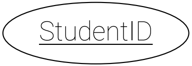


Uniquely identifies a specific entity. Examples: ID, Social Security Number, URL

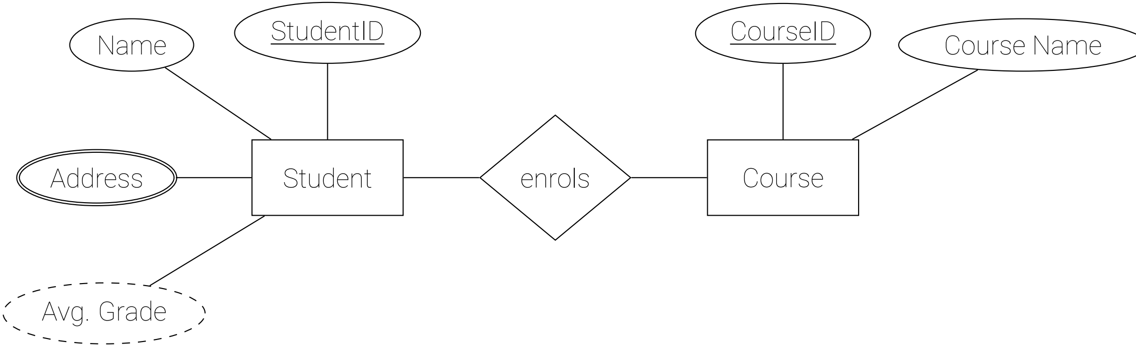
Multivalued attribute

Attribute consisting of multiple values (for Database needs to be split into columns or own table) Examples: Address, Name (First, Last, Title, …),

Derived attribute

Is an attribute calculated from other attributes. Examples.: Avg. Grade, Number of assigned courses, Yearly Income

Example extended



Cardinalities

Define how often a relationship between entities may occur.

Cardinality Types:

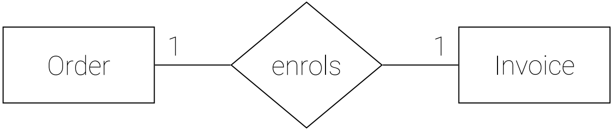
- 1:1 - one to one
- 1:n - one to many
- m:n - many to many

Can be defined more specific with the Min/Max notation:

- 1..1 : 0..n → a department may have 0 to n employees
- 1..1 : 11..11 → one soccer team consists of 11 players

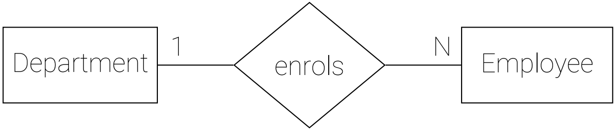
1:1 Cardinality

Direct relationship between two entities.
In a database resolved as a **foreign key** in one (or both) of the tables or both entities are merged into **one table**.



1:N Cardinality

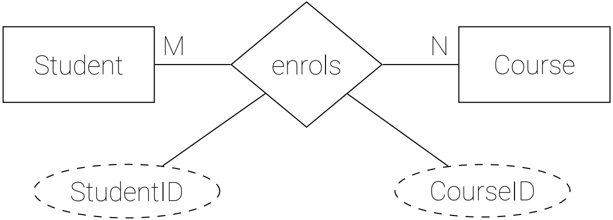
One entity relates to multiple other entities.
In a database resolved as a **foreign key** on the "n" side (e.g. every employee has a department id).



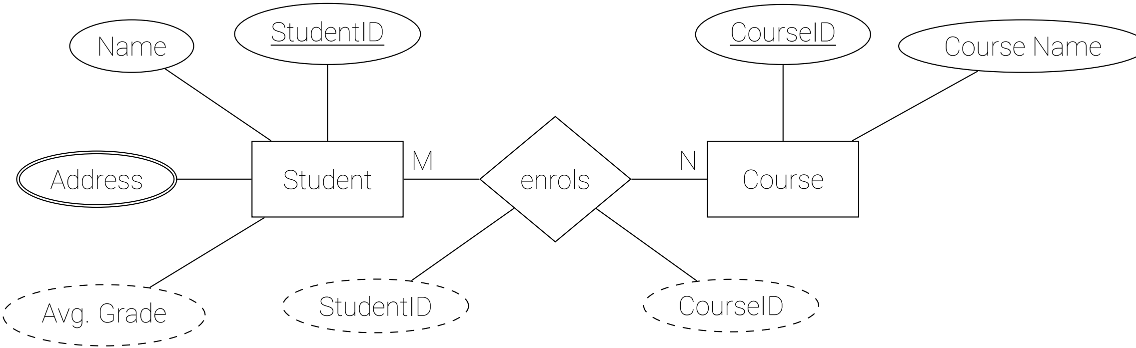
M:N Cardinality

One entity relates to multiple other entities, and vice versa.

In a database resolved as a **new table with two foreign keys**.

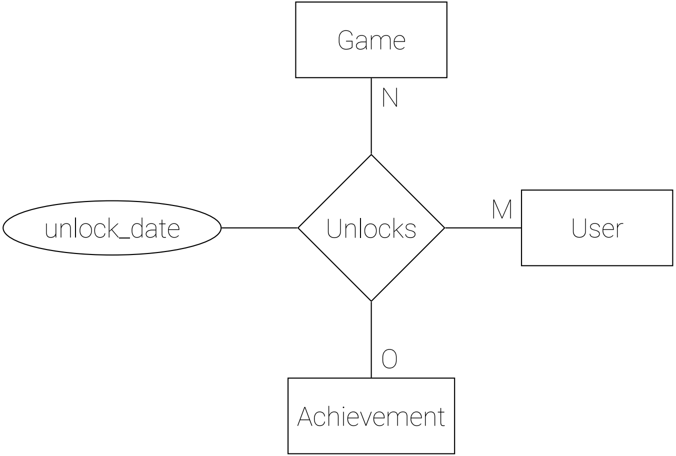


Example with Cardinality



n-ary Relationships

ER diagrams allow relationships between more than one entity (self relationships are also allowed!)



Resolving n-ary Relationships

Depends on the use-case:

- Default → Join Table (also junction table)
  - foreign keys for all tables
  - Composite primary key of all foreign keys
- Resolving relationships
  - 1:N relationship for Game to Achievements
  - M:N for Achievement to User with a Join Table
  - Join required for finding out which games a user has achievements in

Different Notations

ER Diagrams have different notations:

- For Relationships
  - Chen, IDEF1X (see right) are most common
- For Cardinalities
  - on defining side, or accross
  - Expressed in Numbers, or with symbols

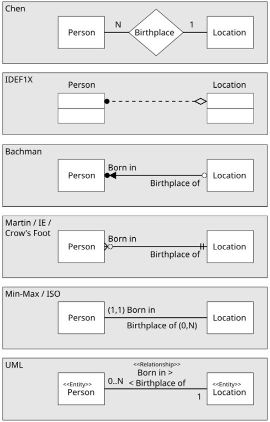
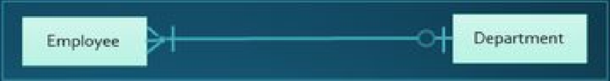


Figure: ER Relationship Representations
Source: Benthompson (https://en.wikipedia.org/wiki/File:ERD_Representation.svg)

Cardinality and Participation for Popular Methodologies

Method	Cardinality	Participation	Example: An employee can work for one department; A department can have one or more employees
Chen	Look-Across	Look-Here	
Crowsfeet	Look Across	Look Across	
I.E.	Look Across	Look Across	
Min/Max	Look Here	Look here	
Min/Max	Look Across	Look Across	

Database Normalization

Is the process of **structuring a relational database** to ensure data **integrity** and **prevent redundancy**.

- Make sure data is not duplicated
- Make sure data integrity is preserved
- Prevent anomalies during Create Update and Delete

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Means that data is correct. E.g. no inconsistencies exist in the database. In relational databases there are the following types of integrity:

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Domain integrity specifies that every column in the database must declare a **domain** (e.g. Integer or List of Values) and be **atomic** (non-decomposable, e.g. Address is split into Street, Number, City, . . .)

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User-defined integrity are **business rules**. For example "A game must have at least 1 achievement per € of its price".

Database Anomalies

Are the reason we need to apply normalization in the first place.

Insertion anomaly occurs when inserting new data, but some data is not available.

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Update anomaly occurs when the database becomes inconsistent on changing data that exists as duplicate in the database.

Deletion anomaly occurs when deltion of data necessitates deletion of additional data that should not be deleted.

Data Anomalies - Example

PublisherID	Publisher Name	Game	Release Date
1	Ubisoft	Assassin's Creed	2007-11-13
1	Ubisoft	Far Cry 3	2012-11-29
2	Electronic Arts	FIFA 21	2020-10-09
2	Electronic Arts	The Sims 4	2014-09-02
3	Nintendo	The Legend of Zelda	1986-02-21

Table: Sample Table Not in Normal Form

Data Anomalies - Insertion Anomaly

PublisherID	Publisher Name	Game	Release Date
1	Ubisoft	Assassin's Creed	2007-11-13
1	Ubisoft	Far Cry 3	2012-11-29
2	Electronic Arts	FIFA 21	2020-10-09
2	Electronic Arts	The Sims 4	2014-09-02
3	Nintendo	The Legend of Zelda	1986-02-21
4	THQ Southic	?	?

Table: Attempting to add THQ Southic leads to an insertion anomaly because the publisher has not yet published a game.

Data Anomalies - Update Anomaly

PublisherID	Publisher Name	Game	Release Date
1	Ubisoft	Assassin's Creed	2007-11-13
1	Ubisoft	Far Cry 3	2012-11-29
2	EA	FIFA 21	2020-10-09
2	Electronic Arts	The Sims 4	2014-09-02
3	Nintendo	The Legend of Zelda	1986-02-21

Table: Updating Electronic Arts to EA leads to an update anomaly since not all occurences were changed.

Data Anomalies - Deletion Anomaly

PublisherID	Publisher Name	Game	Release Date
1	Ubisoft	Assassin's Creed	2007-11-13
1	Ubisoft	Far Cry 3	2012-11-29
2	EA	FIFA 21	2020-10-09
2	Electronic Arts	The Sims 4	2014-09-02
3	Nintendo	**The Legend of Zelda**	1986-02-21

Table: Removing the Legend of Zelda from the catalogue of games also removes Nintendo as a publisher.

Normal Forms

Are definitions that tables need to adhere to in order to **preserve data integrity** and **prevent anomalies**.

0NF	Unnormalized Form
1NF	First Normal Form
2NF	Second Normal Form
3NF	Third Normal Form
EKNF	Elementary-Key Normal Form
BCNF	Boyce-Codd Normal Form
4NF	Fourth Normal Form
ETNF	Essential Tuple Normal Form
5NF	Fifth Normal Form
DKNF	Domain-Key Normal Form
6NF	Sixth Normal Form

0NF Unnormalized Form

The data is not yet normalized. E.g. the data or entity integrity is violated.

PublisherID	Publisher Name	Game	Achievements
1	Ubisoft	Assassin’s Creed	1. Complete the Animus training 2. Collect all feathers
1	Ubisoft	Far Cry 3	1. Liberate all outposts 2. Craft all syringes
2	Electronic Arts	FIFA 21	1. Win 10 matches in FUT 2. Score a hat-trick with any player
2	Electronic Arts	The Sims 4	1. Build a mansion 2. Reach the top of a career
3	Nintendo	The Legend of Zelda	1. Collect all heart containers 2. Defeat Ganon

Table: Table in 0NF, containing duplicate rows (entity integrity) and non-atomic values (domain integrity).

Terms

- **Relationship** A relationship to another table, determined by a foreign key
- **Dependency** The dependencies between columns in a table. Usually written like {Determinant} → {Dependent}
  - **Determinant** is the column on which another column depends. E.g. Game
  - **Dependency** is the column depending on the determinant E.g. MetacriticScore
- **Superkey** is a column (or group of columns) that is used to indentify all attributes in a relationship
- **Candidate Key** a minimal set of columns that can uniquely identify each row
- **Elementary** somnething that is not divisible, e.g. Elementary data.
- **Trivial** Dependencies can be trivial if in {A} → {B} B is part of A.
  - Example: {GameID,GameName}→{GameName} is trivial because the Dependent GameName is part of the Determinant

1NF First Normal Form

Every cell in the table must be **atomic**.

PublisherID	Publisher Name	Game	Achievements
1	Ubisoft	Assassin's Creed	1. Complete the Animus training 2. Collect all feathers
1	Ubisoft	Far Cry 3	1. Liberate all outposts 2. Craft all syringes
2	Electronic Arts	FIFA 21	1. Win 10 matches in FUT 2. Score a hat-trick with any player
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2	Electronic Arts	The Sims 4	1. Build a mansion 2. Reach the top of a career
3	Nintendo	The Legend of Zelda	1. Collect all heart containers 2. Defeat Ganon

Table: Table violates 1NF because Achievements are not atomic. They contain a repeated structure.

Why do we need 1NF?

- Insert Anomaly: Adding a new game without any achievements would require inserting NULL or dummy values in Achievements.
- Update Anomaly: Changing or adding an achievement for a game might require overwriting or restructuring the data manually.
- Delete Anomaly: Deleting the last achievement for a game could inadvertently remove the record for that game entirely.
- Queries, updates, and analyses become inconsistent because the structure doesn’t conform to a fixed schema.
  - If some rows store achievements as lists and others as single strings, it complicates querying and interpretation.

1NF First Normal Form Resolved

Every cell in the table must be **atomic**. This ensures entity integrity.

PublisherID	Publisher Name	Game	Achievement
1	Ubisoft	Assassin's Creed	Complete the Animus training
1	Ubisoft	Assassin's Creed	Collect all feathers
1	Ubisoft	Far Cry 3	Liberate all outposts
1	Ubisoft	Far Cry 3	Craft all syringes
2	Electronic Arts	FIFA 21	Win 10 matches in FUT
2	Electronic Arts	FIFA 21	Score a hat-trick with any player
2	Electronic Arts	The Sims 4	Build a mansion
2	Electronic Arts	The Sims 4	Reach the top of a career
3	Nintendo	The Legend of Zelda	Collect all heart containers
3	Nintendo	The Legend of Zelda	Defeat Ganon

Table: Games with Achievements by Publisher (1NF)

2NF First Normal Form

Every column that is not part of the primary key must **depend on the entire primary key**. (+1NF)

PublisherID (PK)	Publisher Name	Game (PK)	Achievement (PK)
1	Ubisoft	Assassin’s Creed	Complete the Animus training
1	Ubisoft	Assassin’s Creed	Collect all feathers
1	Ubisoft	Far Cry 3	Liberate all outposts
1	Ubisoft	Far Cry 3	Craft all syringes
2	Electronic Arts	FIFA 21	Win 10 matches in FUT
2	Electronic Arts	FIFA 21	Score a hat-trick with any player
2	Electronic Arts	The Sims 4	Build a mansion
2	Electronic Arts	The Sims 4	Reach the top of a career
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PublisherID (PK)	Publisher Name	Game (PK)	Achievement (PK)
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1	Ubisoft	Far Cry 3	Liberate all outposts
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2	Electronic Arts	FIFA 21	Win 10 matches in FUT
2	Electronic Arts	FIFA 21	Score a hat-trick with any player
2	Electronic Arts	The Sims 4	Build a mansion
2	Electronic Arts	The Sims 4	Reach the top of a career
3	Nintendo	The Legend of Zelda	Collect all heart containers
3	Nintendo	The Legend of Zelda	Defeat Ganon

Table: Publisher Name only depends on PublisherID and thus violates the 2NF.

Why do we need 2NF?

- Insert Anomaly: Adding a new publisher without a game would be difficult since Game is part of the primary key.
- Update Anomaly: If the name "Ubisoft" changes, it would need to be updated in multiple rows.
- Delete Anomaly: Deleting the last achievement for a game could inadvertently remove the record for that game entirely.
- Queries, updates, and analyses become inconsistent because the structure doesn't conform to a fixed schema.
    - If some rows store achievements as lists and others as single strings, it complicates querying and interpretation.

2NF Second Normal Form Resolved

PublisherID (PK)	Publisher Name
1	Ubisoft
2	Electronic Arts
3	Nintendo

Table: Publishers Table (2NF)

PublisherID (PK)	Game (PK)
1	Assassin's Creed
1	Far Cry 3
2	FIFA 21
2	The Sims 4
3	The Legend of Zelda

Table: Games Table (2NF)

Game (PK)	Achievement (PK)
Assassin's Creed	Complete the Animus training
Assassin's Creed	Collect all feathers
Far Cry 3	Liberate all outposts
Far Cry 3	Craft all syringes
FIFA 21	Win 10 matches in FUT
FIFA 21	Score a hat-trick with any player
The Sims 4	Build a mansion
The Sims 4	Reach the top of a career
The Legend of Zelda	Collect all heart containers
The Legend of Zelda	Defeat Ganon

Table: Achievements Table (2NF)

2NF Expanded for one Table

Game (PK)	Achievement (PK)	Achievement Type	Points	Achievement Description
Assassin's Creed	Complete the Animus training	Story	50	Complete the initial tutorial
Assassin's Creed	Collect everything	Exploration	30	Find all hidden feathers in the game
Far Cry 3	Liberate all outposts	Exploration	30	Free all enemy outposts on the map
Far Cry 3	Craft all syringes	Crafting	20	Create every type of syringe
FIFA 21	Win 10 matches in FUT	Gameplay	25	Achieve 10 wins in FIFA Ultimate Team
FIFA 21	Score a hat-trick with any player	Gameplay	25	Score 3 goals with the same player in a match
The Sims 4	Build a mansion	Creativity	40	Design and build a luxury mansion
The Sims 4	Reach the top of a career	Progression	35	Max out any career path
The Legend of Zelda	Collect everything	Exploration	30	Find every heart container
The Legend of Zelda	Defeat Ganon	Story	50	Defeat the final boss to save Hyrule

Table: Achievements Table in 2NF with Additional Columns. All columns satisfy 2NF, by depending on an Achievment in a Game, i.e. the entire primary key.

3NF Third Normal Form

Every column that is not part of the primary key depends on the primary key. I.e. there are **no transitive dependencies**. (+2NF)

Game (PK)	**Achievement (PK)**	**Achievement Type**	**Points**	**Achievement Description**
Assassin's Creed	Complete the Animus training	Story	50	Complete the initial tutorial
Assassin's Creed	Collect everything	Exploration	30	Find all hidden feathers in the game
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The Legend of Zelda	Collect everything	Exploration	30	Find every heart container
The Legend of Zelda	Defeat Ganon	Story	50	Defeat the final boss to save Hyrule

3NF Third Normal Form

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Game (PK)	Achievement (PK)	Achievement Type	Points	Achievement Description
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Assassin's Creed	Collect everything	Exploration	30	Find all hidden feathers in the game
Far Cry 3	Liberate all outposts	Exploration	30	Free all enemy outposts on the map
Far Cry 3	Craft all syringes	Crafting	20	Create every type of syringe
FIFA 21	Win 10 matches in FUT	Gameplay	25	Achieve 10 wins in FIFA Ultimate Team
FIFA 21	Score a hat-trick with any player	Gameplay	25	Score 3 goals with the same player in a match
The Sims 4	Build a mansion	Creativity	40	Design and build a luxury mansion
The Sims 4	Reach the top of a career	Progression	35	Max out any career path
The Legend of Zelda	Collect everything	Exploration	30	Find every heart container
The Legend of Zelda	Defeat Ganon	Story	50	Defeat the final boss to save Hyrule

Table: Achievements Table does not satisfy 3NF because Points depend on Achievement Type (which depends on the Achievement), which is a transitive dependency.

Why do we need 3NF?

- Insert Anomaly: Adding a new Achievement Type like Puzzle with its points isn't possible without a corresponding achievement.
- Update Anomaly: If Story achievements are updated from 50 points to 60, you must update every row in the table where Achievement Type is Story. If one row is missed, the data becomes inconsistent.
- Delete Anomaly: If all Story achievements are deleted, the points for Story could also be lost.
- 3NF minimizes redundancy, and makes the database more flexible for data modification.

3NF Third Normal Form Resolved

Game (PK)	Achievement (PK)	Achievement Type	Achievement Description
Assassin's Creed	Complete the Animus training	Story	Complete the initial tutorial
Assassin's Creed	Collect all feathers	Exploration	Find all hidden feathers in the game
Far Cry 3	Liberate all outposts	Exploration	Free all enemy outposts on the map
Far Cry 3	Craft all syringes	Crafting	Create every type of syringe
FIFA 21	Win 10 matches in FUT	Gameplay	Achieve 10 wins in FIFA Ultimate Team
FIFA 21	Score a hat-trick with any player	Gameplay	Score 3 goals with the same player in a match
The Sims 4	Build a mansion	Creativity	Design and build a luxury mansion
The Sims 4	Reach the top of a career	Progression	Max out any career path
The Legend of Zelda	Collect all heart containers	Exploration	Find every heart container
The Legend of Zelda	Defeat Ganon	Story	Defeat the final boss to save Hyrule

Table: Achievements Table in 3NF (No Transitive Dependency)

Achievement Type (PK)	Points
Story	50
Exploration	30
Crafting	20
Gameplay	25
Creativity	40
Progression	35

Table: Achievement Types Table with Points

Elementary Key normal form

Is an enhancement for 3NF that usually only occurs in edge cases. A table is in EKNF if every **non-elementary-key column depends on the entire key** and nothing else. (+3NF)
. . . in simpler terms. Key columns must depend only on the key.

Reviewer (PK)	Game (PK)	Platform	Score
Mark Brown	Assassin's Creed	PC	8.5
Jane Smith	Assassin's Creed	Xbox 360	8.0
Mark Brown	Far Cry 3	PC	9.0
Sarah Lee	FIFA 21	PS4	7.5
Mark Brown	FIFA 21	PC	7.8
Sarah Lee	The Sims 4	PS4	6.9
Mark Brown	The Sims 4	PC	7.2

Table: Game Reviews Table with Reviewer, Game, Platform, and Score

Elementary Key normal form

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Table: The Reviewer depends on the Platform (and vice versa)

Why do we need EKNF?

- EKNF primarily minimizes redundancy even further and reduces maintenance complexity.
- EKNF does prevent anomalies by preventing partial changes
  - Only updating the "Difficulty" of "Story" does not touch the "Points" entry.
- EKNF is rarely needed.

EKNF Form Resolved

Reviewer	Platform (PK)
Mark Brown	PC
Jane Smith	Xbox 360
Sarah Lee	PS4

Table: Reviewer-Platform Table

Game (PK)	Reviewer (PK|FK)	Score
Assassin's Creed	Mark Brown	8.5
Assassin's Creed	Jane Smith	8.0
Far Cry 3	Mark Brown	9.0
FIFA 21	Sarah Lee	7.5
FIFA 21	Mark Brown	7.8
The Sims 4	Sarah Lee	6.9
The Sims 4	Mark Brown	7.2

Table: Game-Reviewer Table

Boyce-Codd Normal Form (BCNF)

Is a different enhancement for 3NF that also only rarely occurs. A table is in BCNF if every functional dependency $X \rightarrow Y$ is **trivial** or X is **a superkey**.
...in simpler terms. **all columns (key or not) must depend on the entire key**.

Track	Session	Channel	Start Time	Session Name
RPG	1	GameChannel1	12:00 PM	Best Storytelling
RPG	2	GameChannel1	1:00 PM	Character Design
Shooter	1	GameChannel2	12:00 PM	Best Gunplay
Shooter	2	GameChannel2	1:00 PM	Multiplayer Mechanics

Table: Game Awards Table

Boyce-Codd Normal Form (BCNF)

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Table: Game Awards Table is not in BCNF because of the overlapping keys {Track, Session};

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Table: Game Awards Table is not in BCNF because of the overlapping keys {Track, Session}; {Channel, StartTime}; {Channel, Session}; {Track, StartTime}

Why do we need BCNF?

- BCNF is stricter than 3NF and makes sure that only superkeys are used.
- BCNF reduces redundancies even further, thus reducing insertion, update or deletion anomalies.
- BCNF could be *unwanted* because it sometimes does not preserve dependencies.
  - see `https://youtu.be/0B6klx8IIH8`

BCNF Form Resolved

Track	Channel
RPG	GameChannel1
Shooter	GameChannel2

Table: Track-Channel Table

Track	Session	Start Time	Session Name
RPG	1	12:00 PM	Best Storytelling
RPG	2	1:00 PM	Character Design
Shooter	1	12:00 PM	Best Gunplay
Shooter	2	1:00 PM	Multiplayer Mechanics

Table: Session Table

Normal Forms - what we won't cover

The following normal forms are also important, but out of scope for this lecture:

4NF	Fourth Normal Form - There should not be multiple dependencies on the primary key that have nothing to do with each other
ETNF	Essential Tuple Normal Form - Joins over tables have a superkey component
5NF	Fifth Normal Form - Joins over tables consist of only superkey components
DKNF	Domain-Key Normal Form - All database constraints are only from domain or keys (no business rules)
6NF	Sixth Normal Form - All tables consist of only one dependency

How far should a database be normalized?

normalize till it hurts, denormalize till it works

- Most times **3NF** is both **sufficient** and **needed to ensure data integrity**.
- For very **complex relationships** or use cases such as data warehousing you might want to **normalize further**.
- For performance you may want to **denormalize** and duplicate data to prevent joins. Especially important for Big Data.

Data Modelling process

1. Identify the **entities** in your software
2. Identify the **relationships**
3. Identify **keys** and **attributes**
4. **Normalize** as far as needed
5. Check your **performance** and **queries**